An Interactive Visualization Tool for Transmission Clustering Across All TN93 Thresholds

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Abstract

Epidemiologists use transmission clustering based on genetic sequences to analyze the spread of disease. As the collection of viral sequences has become easier and less expensive, molecular epidemiology efforts are rapidly increasing in scale. HIV-TRACE, a standard tool in viral molecular epidemiology, performs transmission clustering by estimating pairwise distances between viral sequences under the TN93 model of evolution and linking individuals whose pairwise distances are below a chosen distance threshold. The choice of distance threshold is challenging; it varies between viruses, populations, and regions of the viral genome. As a result, epidemiologists often perform transmission clustering using multiple distance thresholds, but it is unclear what specific thresholds should be selected. Here, we present TreeN93, a web-based tool that enables studying transmission clustering results across all possible thresholds. Given pairwise distances, TreeN93 constructs an ultrametric single-linkage hierarchical clustering tree; cutting the resulting tree *d* distance above the leaves yields the transmission clusters obtained using threshold *d*. TreeN93 allows the user to visualize and interact with the clustering tree, and users can easily view and export clusters at any selected distance threshold. TreeN93 is available at https://moshiri-lab.github.io/TreeN93/, and the code is available at https://github.com/Moshiri-Lab/TreeN93.

Introduction

Current Landscape

- Modern epidemiology relies on transmission clustering of viral sequences to slow disease transmission
- Challenge: how to determine clusters
- Clusters chosen based on genetic differences, quantified using TN93 distance
- Distance threshold chosen for clustering for a given virus or population

Our Solution

- Web-based tool for analyzing transmission clusters across all possible distance thresholds
- Constructs an ultrametric single-linkage hierarchical clustering tree from TN93 distances
- Clusters can be viewed and saved at any user-chosen threshold

Methods

Build:
A tool to build a tree from pairwise TN93 distances between leaf nodes, and to save the tree in the Newick file format.

- Pairwise distances inputted as a CSV
- Option to input pairwise distances in list or matrix format
- Tree created using an implementation of Disjoint Set data structure

Viz:
A tool to visualize clusters on the tree from a user-inputted distance threshold, and to find the minimum, threshold that yields the maximum number of clusters.

- Tree inputted as a Newick file
- User changes threshold by typing a number, moving a slider, or selecting any node on the tree
- Colored tree can be saved as a png or svg file and clusters can be saved as a text file
- Control panel to change size and shape of tree for optimal viewing
- "Maximize" button sets the threshold to the minimum threshold that yields the maximum clusters
- Clusters re-determined when threshold changes
- Clusters visualized by coloring of branches and nodes

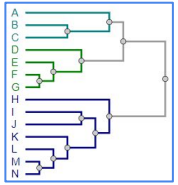


Figure 1: Example Clustering Tree

Results - Webpages

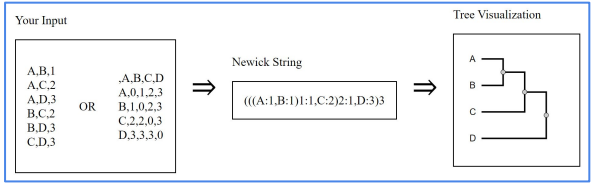
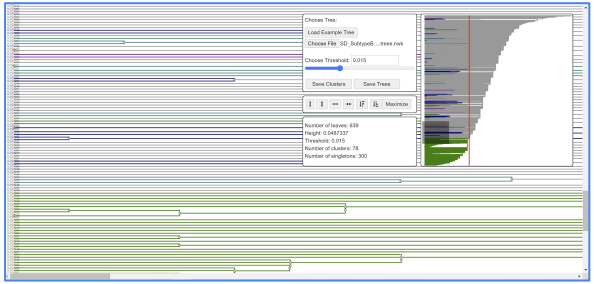


Figure 2: Steps in the Build Process ↑

Figure 3: Full Window of Viz Page ↓



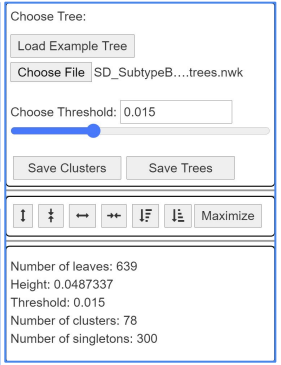


Figure 4: Control Panel of Viz Page

Results - Clustering Trees

Figure 5: HIV Subtype B from San Diego, Clusters Maximized

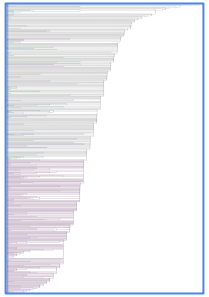


Figure 6: HIV Subtype B from San Diego, Threshold Set to 0.2

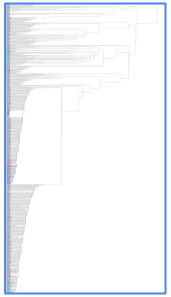


Figure 7: Hepatitis C from Los Alamos, Clusters Maximized

Figure 8: Hepatitis C from Los Alamos, Threshold Set to 0.5

Conclusion

- TreeN93 is a web-based tool for building and visualizing clustering trees
- Build constructs trees from pairwise TN93 distances
- Viz visualizes the clusters at any distance threshold
- Useful for known viruses to refine existing clustering threshold distances
- Useful for novel viruses to examine a range of thresholds and choose the best for future study

References

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